

SWIR oblique-scanning scatterometry using global sensitivity analysis and back focal plane imaging for hidden HAR microstructure critical dimension measurement

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ABSTRACT

Advancements in 3D integrated circuits (3DIC) and Chip-on-Wafer-on-Substrate (CoWoS) packaging demand precise metrology of high aspect ratio (HAR) microstructures, challenging conventional inline measurement techniques. Traditional back focal plane (BFP) scatterometry offers high sensitivity to lateral critical dimensions (CDs) but lacks depth resolution. To address this, we present a hybrid visible-to-shortwave infrared (SWIR) spectroscopic scatterometry system that combines BFP scatterometry with spectroscopic reflectometry. A low numerical aperture (NA) normal incidence configuration and angle-scanning module are employed to enhance depth sensitivity and lateral CD accuracy, respectively. The SWIR range improves bottom CD and sidewall angle assessment due to deeper penetration. Global sensitivity analysis (GSA) guides system optimization, and BFP imaging supports lateral CD extraction. Experimental validation demonstrates CD measurement errors under 1.95%, confirming the system's precision for HAR metrology.

Keywords: High Aspect Ratio Structure, Scatterometry, Hybrid Metrology, Back Focal Plane, Global Sensitivity Analysis.

1. INTRODUCTION

To address the increasing demands for quality and process control in HAR microstructures, this study proposes an innovative optical metrology system and establishes a hybrid measurement strategy. By leveraging the complementary sensitivities of different measurement techniques, the overall measurement capability is enhanced, thereby reducing the ill-posedness of inverse reconstruction[1-5].

2. METHODOLOGY

2.1 Global sensitivity analysis & Hybrid metrology

This study employs GSA to quantify the influence of various critical dimensions on different optical responses in HAR structures[6-8], thereby establishing a hybrid metrology strategy[9-11]. Spectroscopic reflectometry is the primary tool, while BFP scatterometry is a secondary tool that provides high sensitivity to lateral dimensions such as TCD and spacing. The proposed method enhances the accuracy and stability of inverse reconstruction by implementing a feedforward mechanism.

2.2 Surrogate model & Optical system design

This study constructed a Gaussian Process Regression (GPR) surrogate model[12, 13], to comprehensively evaluate sensitivity distributions under different system configurations to optimize the overall design and measurement configuration. The surrogate model enabled the selection of incident angle and polarization combinations with the highest measurement sensitivity. The experimental system (as shown in Fig. 1) integrates a VIS-SWIR light source, a spectroscopic reflectometry module, an incident angle scanning module, and a BFP imaging module, enabling multimodal, non-destructive measurement of HAR structures[14, 15].

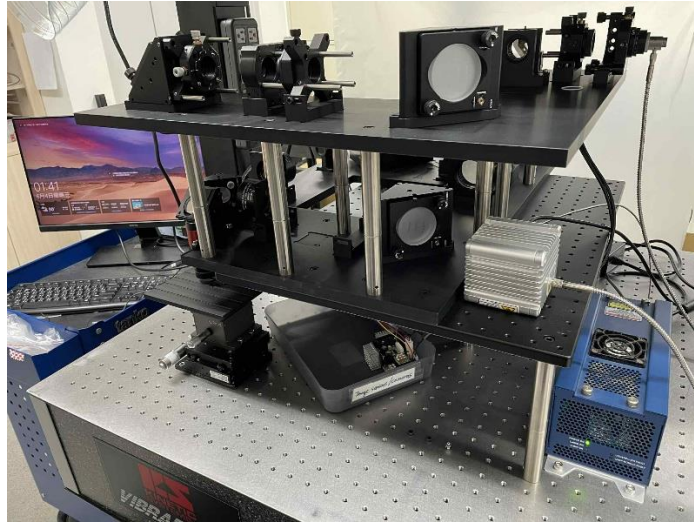


Figure 1 The developed optical measurement system.

3. EXPERIMENT RESULTS & DISCUSSION

3.1 Optimizing Measurement Strategies via GPR Surrogate Model

RCWA electromagnetic simulations were performed with 35 combinations of incident angles and polarization states for the Si trench structure. Surrogate models (as shown in Fig. 2) were constructed based on the zero-order spectrum and the intensity ratios of the zero and ± 1 st-order diffractions to determine the optimal measurement configuration.

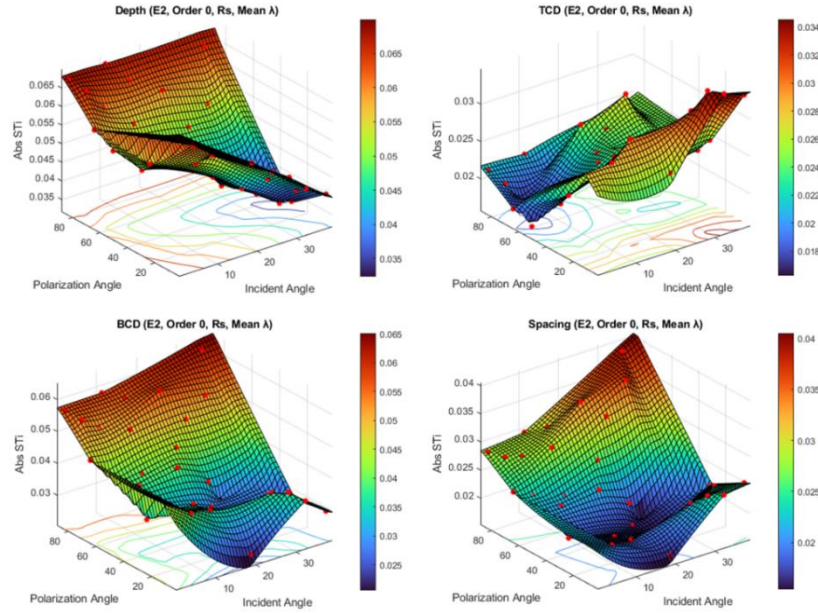


Figure 2 Schematic diagram of Surrogate Models.

Additionally, this study utilized Lumerical FDTD simulations to extract Poynting vectors and compare the propagation characteristics of visible and infrared wavelengths in the Si trench (as shown in Fig. 3). The results indicate that visible light is mainly confined to the air regions between the structures, with limited interaction with the silicon substrate. In contrast, SWIR wavelengths penetrate the silicon and reach deeper into the structure, enabling more significant interactions. Therefore, the SWIR band demonstrates greater potential for resolving deep critical dimensions.

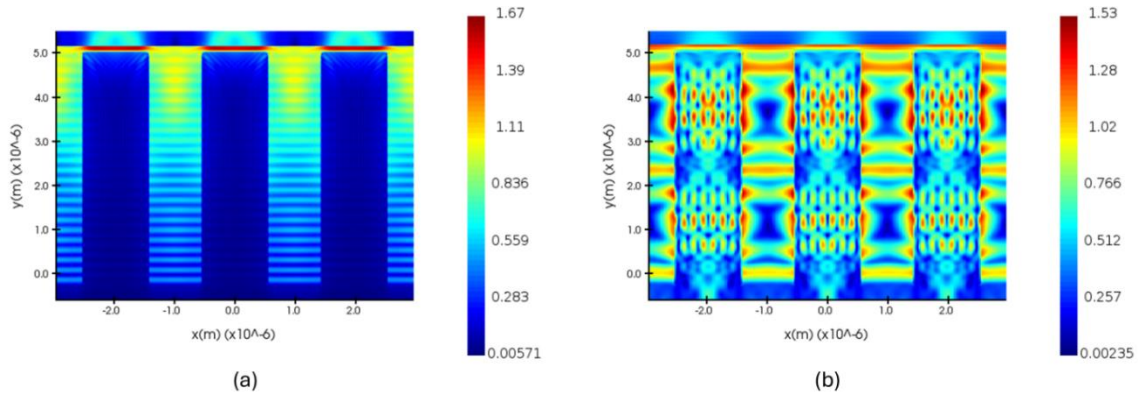


Figure 3 Poynting vectors for Si Trench: (a) using VIS wavelength range and (b) using SWIR wavelength range.

3.2 Critical Dimension Measurement of Si Trench Structures

This section conducts hybrid measurement implementation and validation on the Si trench (as shown in Fig. 4), using the incident angle and polarization state with the highest sensitivity identified by the surrogate model. The goal is to verify the proposed system's measurement accuracy and stability for HAR structures.

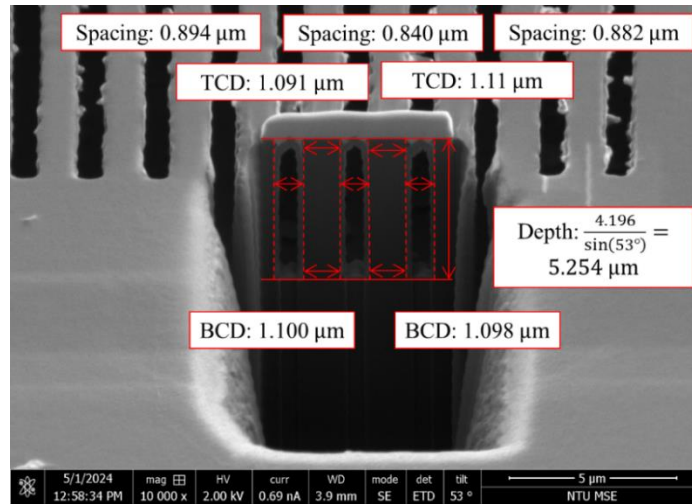


Figure 4 Si Trench structure and SEM measurement result.

The proposed strategy effectively integrates different signal modalities. It uses the BFP measurement results as the initial point for simulated annealing, thereby improving the accuracy and stability of the inverse reconstruction. The measurement results for both SWIR and visible wavelength ranges are shown in Table 1.

Table 1 Comparison of Simulated Annealing Results under SWIR(700 ~ 1100 nm) and VIS(450 ~ 700nm) Wavelength.

	Depth(μm)	TCD(μm)	BCD(μm)	Spacing(μm)
SEM	5.254	1.101	1.099	0.872
SWIR result	5.304	1.1055	1.106	0.889
VIS result	5.549	1.096	1.114	0.858
SWIR Bias(%)	0.95%	0.41%	0.64%	1.95%
VIS Bias(%)	5.61%	-0.45%	1.36%	-1.61%

As a result, SWIR enhances the measurement accuracy of the BCD, with all CD measurement bias remaining within 2% compared to SEM results. Moreover, by incorporating BFP measurements and using diffraction angle constraints to limit the solution space for lateral dimensions, the dimensionality of the simulation database is reduced. This effectively improves the depth measurement process and enhances the convergence efficiency of the simulated annealing algorithm, leading to higher-precision inverse reconstruction.

4. CONCLUSIONS

In the experiment, combining the optimized configuration with the hybrid measurement strategy and simulated annealing, the measurement bias for all CDs is below 2%. Notably, compared to visible light, SWIR exhibits stronger penetration and interaction capabilities, making it particularly suitable for analyzing bottom or hidden structural features. This study establishes a sensitivity-guided hybrid metrology framework with high resolution and stability. Future work will extend its application to a broader range of structures and materials while continuously improving its practicality and versatility.

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